COHERENT MODULES ON PROJ OF A GENERAL NOETHERIAN GRADED RING: FOR A NOETHERIAN GRADED AND X = PROJ(A), THE FUNCTOR M -> M^TILDE INDUCES AN EQUIVALENCE MOD^FG_A / MOD^FG_{A,IRRELEVANT} -> COH(O_X) WHOSE QUASI-INVERSE IS F |-> BIGOPLUS_{N >= 0} GAMMA(X, F(N)).

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4442-4453 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 4455-4527, the complete author proof (71 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Full derivation in span. Depends on lemma-recover-tail-graded-module-general and lemma-coherent-on-proj-general (same chapter; the former is installed in this wave as L16356).

Editorial note. Span computed from the \label marker through the enclosing \end{proposition}; precard_gate RESULT=OK (probe L9999).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original proposition 01M7: sources/stacks-coherent/pinned-original-source.tex, source lines 4527–4527. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 02MQ: sources/stacks-coherent/pinned-original-source.tex, source lines 4527–4527. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 02MS: sources/stacks-coherent/pinned-original-source.tex, source lines 4527–4527. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B5Q: sources/stacks-coherent/pinned-original-source.tex, source lines 4237–4255. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B5R: sources/stacks-coherent/pinned-original-source.tex, source lines 4298–4313. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0AG5: sources/stacks-coherent/pinned-original-source.tex, source lines 4527–4527. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0BXD: sources/stacks-coherent/pinned-original-source.tex, source lines 4168–4180. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let A be a Noetherian graded ring. Set $X = Proj(A)$. The functor $M \mapsto \widetilde{M}$ induces an equivalence*

$$Mod_A^{fg}/Mod_{A,irrelevant}^{fg} \longrightarrow Coh(\mathcal{O}_X)$$

whose quasi-inverse is given by $\mathcal{F} \longmapsto \bigoplus_{n\geq 0} \Gamma(X, \mathcal{F}(n))$.

Proof. We urge the reader to read the proof in the case where A is generated in degree 1 first, see Proposition 0BXD. Let $f_1, \ldots, f_r \in A$ be homogeneous elements of positive degree which generate A over A_0. Let d be the lcm of the degrees d_i of f_i. Let M be a finite A-module. Let us show that $\widetilde{M}$ is zero if and only if M is an irrelevant graded A-module (as defined above the statement of the proposition). Namely, let $x \in M$ be a homogeneous element. Choose $k \in \mathbf{Z}$ sufficiently small and let $N \to N'$ and $M \to N'$ be as in Lemma 0B5R. We may also pick l sufficiently large such that $M_n \to N_n$ is an isomorphism for $n \geq l$. If $\widetilde{M}$ is zero, then $N = 0$. Thus for any $f \in A_+$ homogeneous with $\deg(f) + \deg(x) = nd$ and $nd > l$ we see that fx is zero because $N_{nd} \to N'_{nd}$ and $M_{nd} \to N'_{nd}$ are isomorphisms. Hence x is irrelevant. Conversely, assume M is irrelevant. Then M_{nd} is zero for $n \gg 0$ (see discussion above proposition). Clearly this implies that $M_{(f_i)} = M_{(f_i^{d/\deg(f_i)})} = 0$, whence $\widetilde{M} = 0$ by construction.

It follows that the subcategory $\mathrm{Mod}_{A,irrelevant}^{fg}$ is a Serre subcategory of Mod_A^{fg} as the kernel of the exact functor $M \mapsto \widetilde{M}$, see Homology, Lemma 02MQ and Constructions, Lemma 01M7. Hence the quotient category on the left of the arrow is defined in Homology, Lemma 02MS. To define the functor of the proposition, it suffices to show that the functor $M \mapsto \widetilde{M}$ sends irrelevant modules to 0 which we have shown above.

By Lemma 0B5Q the proposed quasi-inverse makes sense. Namely, the lemma shows that $\mathcal{F} \longmapsto \bigoplus_{n\geq 0} \Gamma(X, \mathcal{F}(n))$ is a functor $Coh(\mathcal{O}_X) \to \mathrm{Mod}_A^{fg}$ which we can compose with the quotient functor $\mathrm{Mod}_A^{fg} \to \mathrm{Mod}_A^{fg}/\mathrm{Mod}_{A,irrelevant}^{fg}$.

By Lemma 0B5R the composite left to right to left is isomorphic to the identity functor. Namely, let M be a finite graded A-module and let $k \in \mathbf{Z}$ sufficiently small and let $N \to N'$ and $M \to N'$ be as in Lemma 0B5R. Then the kernel and cokernel of $M \to N'$ are nonzero in only finitely many degrees, hence are irrelevant. Moreover, the kernel and cokernel of the map $N \to N'$ are zero in all sufficiently large degrees divisible by d, hence these are irrelevant modules too. Thus $M \to N'$ and $N \to N'$ are both isomorphisms in the quotient category, as desired.

Finally, let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Set $M = \bigoplus_{n\in\mathbf{Z}} \Gamma(X, \mathcal{F}(n))$ viewed as a graded A-module, so that our functor sends $\mathcal{F}$ to $M_{\geq 0} = \bigoplus_{n\geq 0} M_n$. By Properties, Lemma 0AG5 the canonical map $\widetilde{M} \to \mathcal{F}$ is an isomorphism. Since the inclusion map $M_{\geq 0} \to M$ defines an isomorphism $\widetilde{M_{\geq 0}} \to \widetilde{M}$ we conclude that the composite right to left to right is isomorphic to the identity functor as well. □